Instructions & Defense Runbook

How to set up, verify, regenerate, and defend every number in the submission · repo: `github.com/Manureddy148/AI_Task2` · compiled 2026-09-04

1 Setup

```
git clone https://github.com/Manureddy148/AI_Task2.git && cd AI_Task2
pip install -r submission/requirements.txt        # Python 3.10+; no torch, no GPU
```

Network is needed once: tokenizer files (tiktoken/HuggingFace) and, only if you rebuild the corpus, the public FLORES-200 tarball (24 MB). **Layout:** scripts find the audited inputs via `submission/_paths.py` — the committed `starter_kit/` works as-is; any other layout: set `STARTER_KIT=<path containing bench/>`.

2 Verify first (the 30-second credibility check)

```
python submission/verify.py            # expected final line:  ALL 50 NUMBERS VERIFIED
python submission/verify.py --full     # also regenerates results/romanization/reconcile first (~3 min)
```

The 50 assertions cover: byte-exact REPORT_v0 reproduction, each bug's isolated delta, corpus SHA-256 pins, the full parity grid, every derived ratio quoted in prose, romanization on both tokenizer generations, all 13 bench-log reconciliations, and the roofline fit. CI runs the same harness on every push.

Defense preparation: `python submission/defense_drill.py` walks the candidate's personal checklist interactively (~10 min) — runs verify, the A0 repro, a live corpus toggle, B1 on paper, and one bench row by hand, showing expected vs actual at each step.

3 Regenerate everything from scratch

```
python submission/partA/corpus/build_corpus.py     # corpus (downloads on first run; hash-pinned)
python submission/partA/audit/experiments.py       # A2 one-toggle bug isolation
python submission/partA/fertility_v1.py            # A3 grid: 5 tokenizers × 4 denominators × 7 languages
python submission/partA/audit/romanization.py      # romanization + Tamil artifact bound + probes
python submission/partB/kv_math.py                 # B1 arithmetic
python submission/partB/reconcile.py               # 13-row replay, goodput, roofline, fp8 forecast
python submission/figures/make_figures.py          # report figures (annotations computed from data)
python submission/verify.py                        # assert it all again
```

4 Live-defense overrides ("change this input right now")

Ask	Command
Paste a modified toy corpus into A2	`python submission/partA/audit/experiments.py --hin pasted.txt` — every variant, the data facts and the bootstrap CI move with it (sizes are derived, not hardcoded).
Swap one language's file in A3	`python submission/partA/fertility_v1.py --corpus hin=pasted.txt` (repeatable; line-alignment with eng is asserted with a clear error).

Point Part B at an edited bench log	`python submission/partB/reconcile.py --log edited.csv` — all seven sections recompute; row binding is unique-match asserted.
Change a spec constant	Edit the constant in `kv_math.py` — every printed equation string is an f-string over the constants; nothing numeric is retyped literal text.
Different starter-kit location	`STARTER_KIT=<path>` env var.

5 Defense crib — the numbers and where each one lives

Number	Derivation	Source of truth
114,688 B/token KV (112 KiB)	2 (K,V) × 28 layers × 8 KV heads × 128 head_dim × 2 B fp16	partB/kv_math.py
25.7 → ~25 concurrent 4k seqs	(0.92·24 − 8.4 − 1.6) GB ÷ (4096 × 114,688 B)	kv_math.py; log confirms: util 0.93@24, preempt 7=32−25, 23=48−25
201 tok/s honest goodput	512·24/61.16 by definition; independent route n/itl = 249.8 tok/s over the 49.1 s decode phase; counter/8 is an identity, not evidence	reconcile.py §3; capacity.md B3
65% MBU roofline	itl = (8.4 GB + resident·avg_ctx·112 KiB)/(0.65·300 GB/s) fits 13/13 ITLs ±2.8%; compute 1.7 ms vs 63.2 ms memory → memory-bound ~38×	reconcile.py §6
Parity: hin 7.42× · kan 13.59× · tam 15.54× · tel 12.97× · ben 9.61× · mar 7.86×	total tokens ÷ English tokens over the same 1012 parallel sentences (gpt2 = v0's tokenizer); CIs ≤ ±1.28%	partA/results/results.csv; fig1
MuRIL 1.00–1.20×	Same corpus, Indic-focused vocab; unk ≤ 0.38% (integrity column)	results.csv
Romanized hin 2.25× (gpt2) / 1.87× (o200k)	IAST-derived chat approximation; organic-convention probe 2.11/1.76; Tamil is a range 2.37–3.09× (aspirate artifact bound)	partA/audit/romanization_output.md
Bug deltas +0.59 / +2.92 / +0.35%	one-toggle isolation vs byte-exact v0 baseline 5.8871×	partA/audit/experiment_output.md

6 The five hardest defense questions — and where the answer is

- **"Why is tokens-per-word wrong? It's the standard fertility metric."** Fine *within* a language; across languages it errs −15% (hin) to +36% (kan) against content-constant truth because words carry different content per language — findings.md C1 table, derived from results.csv.
- **"Your two metrics agreed — doesn't that confirm the result?"** Shared numerator ⇒ correlation by construction; the repo shows the same tokens giving 6.34× (word) / 11.39× (grapheme) / 2.90× (byte) / 7.46× (code point) — findings.md C2 plus results.md's denominator section. (And the audit applied this to itself: see B3's relabelled "identity".)

- **"Is 7.4× what production pays?"** Unknown — the spec's 128k vocab is not gpt2; the premium is bracketed (cl100k row is scale-illustration) until recommendation 6 measures the real tokenizer. REPORT_v1 §4.
- **"Why does throughput fall at batch 32?"** 448 MiB/seq × 32 > 12.08 GB budget → 7 evictions with full re-prefill (~3.5 s trigger) + serialized tail past the 25-slot ceiling; util pegs at 0.97 — capacity.md B2, every column reconciled.
- **"What would falsify your fixes?"** Stated in advance: max_num_seqs=24 → 122.3 s / 0 preemptions at offered 48; fp8-KV → 51 slots, ~2× decode (capacity half is arithmetic-certain, ITL half assumes MBU transfers); counter to pull: vllm:num_preemptions_total ≥7/≥23 — capacity.md B2/B4.

7 Troubleshooting

- **HF rate-limit warning** on first tokenizer download is benign; set `HF_TOKEN` to silence.
- **Corpus hash FAIL after a fresh clone** would mean newline translation — the repo ships `.gitattributes * -text` precisely to prevent it; check that file survived any re-zip.
- **verify.py --full failure**: the failing child's stderr prints through (stdout is suppressed); fix the cause it names.
- Windows/macOS/Linux all fine; everything is CPU-only.

Regenerate these PDFs: `powershell docs/make_pdfs.ps1` (headless Edge/Chromium print of docs/src/*.html).